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Education

Universidade do Porto <i>Master's Degree in Informatics and Computing Engineering</i> • Relevant Coursework: Digital Game Design and Development, Virtual and Augmented Reality, Interactive Graphics Systems, Computer Vision, Artificial Intelligence, Large-Scale Distributed Systems	Sep 2024 - Expected Jul 2026 Porto, Portugal
Universidade do Minho <i>Bachelor's Degree in Informatics Engineering</i> • Graduated with strong focus on software development, computer systems, and algorithms	Sep 2022 - Jul 2024 Braga, Portugal
Universidade do Minho <i>Bachelor's Degree in Biomedical Engineering</i> • Multidisciplinary background that combines programming fundamentals, electronics, mechanics, materials, and biology	Sep 2020 - Jul 2024 Braga, Portugal

Experience

Mindera <i>Master Thesis Intern</i> • Master's Thesis: "External Agents for Human-Like Play in Two-Player Games: A Case Study in 2D Platformers" • Developed an internal project aimed at creating an AI agent capable of acting as a second player in two-player video games. • Engineered the AI to prioritize human-like behavior and believability during gameplay, optimizing for realistic interactions rather than flawless performance. • Integrated computer vision and AI models to process in-game environments in real-time, utilizing NVIDIA NitroGen, YOLO, and Grounding DINO.	Feb 2026 - Jun 2026 Porto, Portugal
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Projects

Escape do Saber Unity • Developed an educational Virtual Reality escape room for the Meta Quest 2 as part of the Virtual and Augmented Reality course. • Designed and implemented 3 subject-related educational mini-games focused on color mixing mechanics, anatomy (skeleton assembly), and English language comprehension. • Created an interactive ending where users complete the experience by collecting a graduation cap.	Repo Link
Balloon Popping Robot Competition Unity, ML-Agents • Engineered a MARL competition in Unity using ML-Agents to simulate an autonomous team-based Battle Royale tournament. • Applied POCA and Curriculum Learning to train robot agents to navigate asymmetric obstacles and adapt to shrinking arena constraints. • Equipped agents with Ray Perception Sensors for real-time obstacle avoidance, offensive tactics, and resource collection.	Repo Link
Right Under Your Moustache Godot • Co-developed a 2D puzzle-stealth game in Godot as part of the Digital Game Design and Development course. • Programmed core gameplay mechanics, including player special abilities, puzzle logic, and enemy detection systems. • Designed original visual assets, character models, background elements, and integrated immersive audio and sound effects.	Repo Link
Races Unity, Vuforia AR • Developed a mobile Augmented Reality (AR) racing game for the Virtual and Augmented Reality course using Vuforia. • Programmed core mechanics, including an AI opponent with selectable difficulties, motion-based controls, and a 6-scene UI flow. • Designed a strategic phase allowing players to dynamically place obstacles and power-ups on the track using AR markers.	Repo Link

Skills

Technical Skills

Game Dev: Godot, Unity
Backend: Python, C, C#, C++ Java, Haskell, .NET
IA & Machine Learning: Unity ML-Agents, NitroGen, YOLO, Grounding DINO
Frontend: JavaScript, React, Vue.js, HTML/CSS, Pug
Mobile: Kotlin, React Native, Flutter
Databases: MongoDB, MySQL, Redis, SQL Server, PostgreSQL, Cassandra, Neo4j, HBase
Tools: Markdown, L^AT_EX, Figma, Docker, Visual Paradigm, SolidWorks, Matlab, Microsoft Office

Languages

Portuguese: Native
English: Fluent
Spanish: Basic

Leadership & Volunteering

Vice-president of Literatuna – University of Minho <i>Student Organization Leadership</i> Coordinated activities and performances of a university musical group.	2023 – 2024 Braga, Portugal
Erasmus+ Project – ANIMA 2022 <i>International Sociocultural Event</i> International meeting of sociocultural animation.	2022 Portugal / Europe
Volunteer – Extracurricular Classes <i>Volunteering</i> Taught extracurricular classes to children aged 6–9, promoting environmental awareness through educational activities.	2022 Braga, Portugal
Summer School of Physics – FCUP <i>Academic Participation</i> Course in relativity, quantum mechanics and nanotechnology held by FCUP.	2019 Porto, Portugal
Erasmus+ Project – Differences – Survival Camp <i>Youth Inclusion Project</i> Support to reintegration of young people with disabilities.	2018 Gerês, Portugal